

SIT330-770: Natural Language Processing

Week 0 - Course Overview

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DEAKIN
UNIVERSITY

- Introduction to SIT₃₃₀₋₇₇₀
 - Unit team
 - Unit objectives and Introduction to NLP
 - Unit structure
 - Teaching and Assessment
 - Unit logistics



- **Unit chair:**
 - *Reda Bouadjenek*
 - Email: reda.bouadjenek@deakin.edu.au



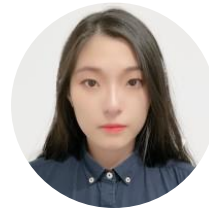
- **Burwood seminars:**
 - *Ngoc Dung Huynh*
 - Email: ndhuynh@deakin.edu.au



- **Online seminar:**
 - *Aymen Khouas*
 - Email: s223766761@deakin.edu.au



- **Online seminar:**
 - *Saeedeh Javadi*
 - Email: saeedeh.javadi@deakin.edu.au



- **Online seminar:**
 - *Echo Zhou*
 - Email: echo.zhou@deakin.edu.au



- **Marking tutor:**
 - *Lakpa Tamang*
 - Email: ld.tamang25@gmail.com

- Dr. Reda Bouadjenek <reda.bouadjenek@deakin.edu.au>
 - Unit chair, Waurn Ponds
- Senior Lecturer of Applied AI at Deakin University since November 2019
 - Interested in information Retrieval, Natural Language Processing, social media analysis, recommender systems
- Previously:
 - **2017-2019**: Research fellow at Toronto University, Canada
 - **2015-2017**: Research fellow at Melbourne University
 - **2014-2015**: Research fellow at inria, France
 - **2009-2013**: PhD student at the University of Paris-Saclay, France

Understanding principals behind Natural Language Processing

What is natural language processing?

The study of language and linguistic interactions from a computational perspective, enabling the development of algorithms and models capable of (a) **natural language understanding (NLU)** and (b) **natural language generation (NLG)**

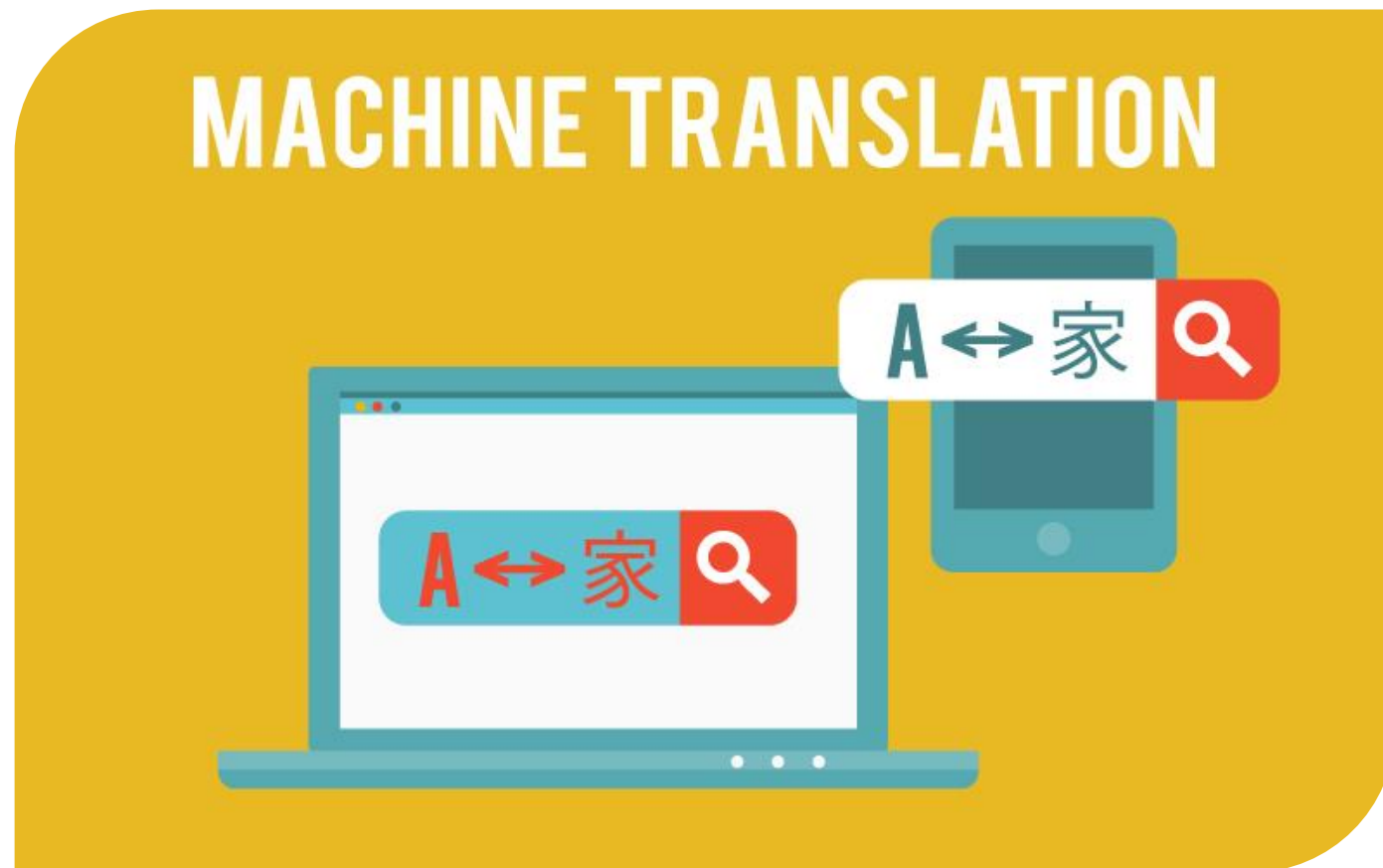


Go beyond the keyword matching

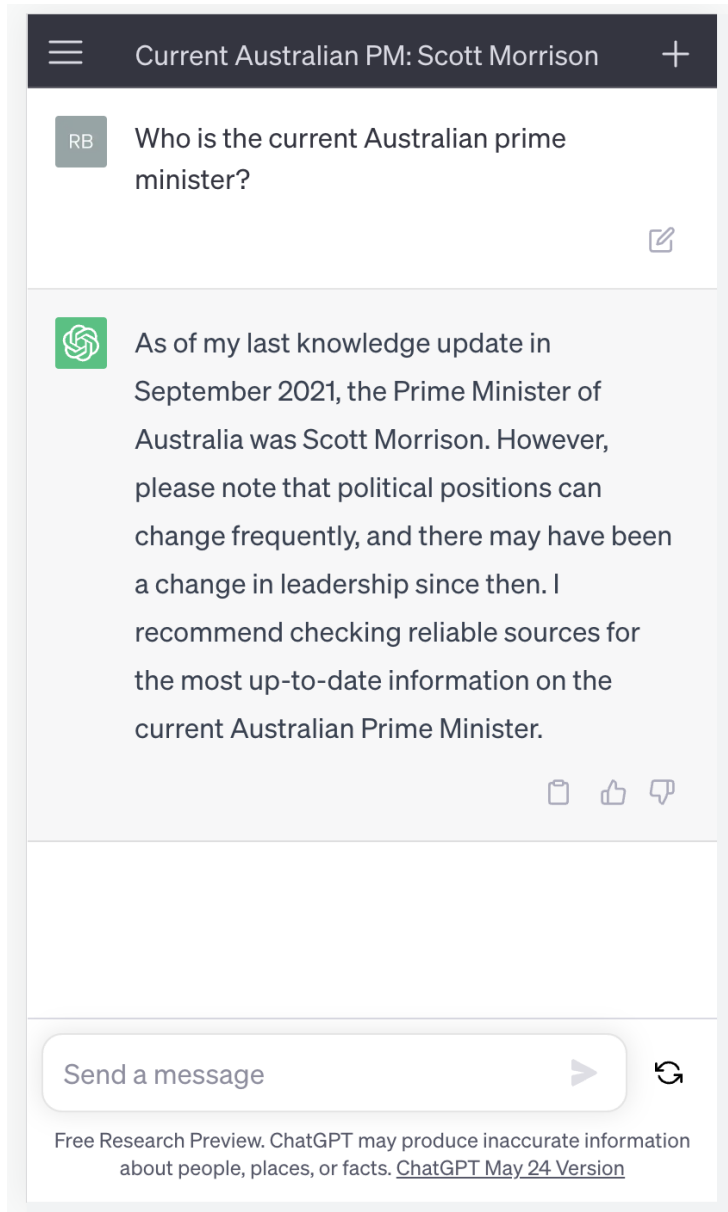


- Identify the **structure** and **meaning** of **words, sentences, texts** and **conversations**
- **Deep** understanding of **broad** language
- NLP is all around us

Machine translation



Conversational System



The ChatGPT logo, consisting of a stylized interlocking knot icon, is positioned to the left of the text "ChatGPT". The background features a light green gradient with a network diagram of nodes and connecting lines.

Sentiment/Opinion Analysis



Negative

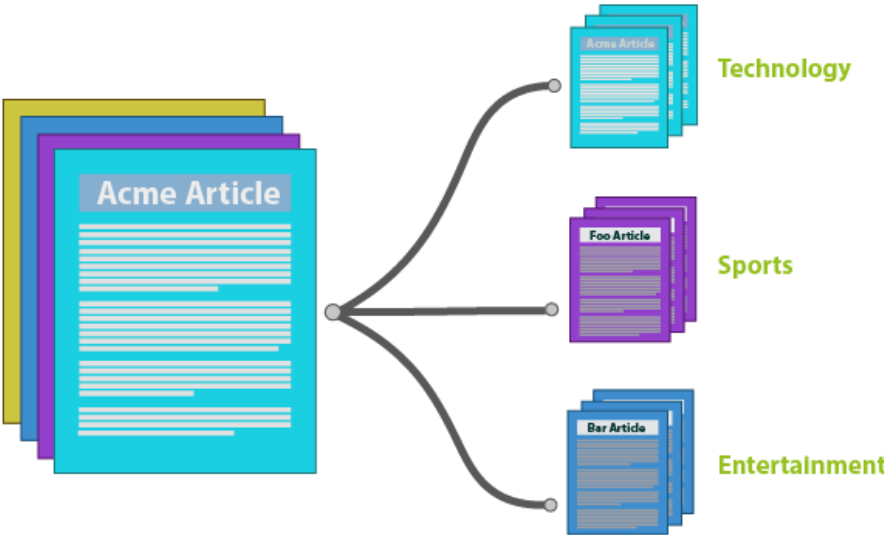


Neutral



Positive

Text Classification



1-50 of 12,404

Primary

Promotions

Social

Updates

Forums

Short paper

SIGIR'24 submission 8282 update - Dear authors, we acknowledge that we received new files for your SIGIR'24 submission. The information about this update is shown below. Num...

Feb 9

Short paper 2

SIGIR'24 submission 8300 update - paper, version 2 (474368 bytes)

Feb 9

Short paper 2

SIGIR 2024 short papers: authorship, COIs and anonymity - Dear Mohamed Reda Bouadjenek, We look forward to your SIGIR 2024 short paper submission, deadline **Feb 8th (Thu...

Feb 7

Resource paper & Re.

SIGIR'24 submission 3233 - Dear authors, We received your submission to SIGIR'24 (The 47th International ACM SIGIR Conference on Research and Development in Information Retr...

Feb 1

IJCNN 2024 4

[IJCNN 2024] Information about paper #1570994543 (A Mask-Based Prior Knowledge Dissemination Method for Deep Neural Networks) has been changed - Author Reda Bouadje...

Jan 31

Natural language instruction



Will it rain tomorrow?

Set an alarm at 8am

Play music by Bruno Mars

How many teaspoons
are in a tablespoon?

Add soap to my
shopping list

Play my *dinner party* playlist

When is Australia Day?

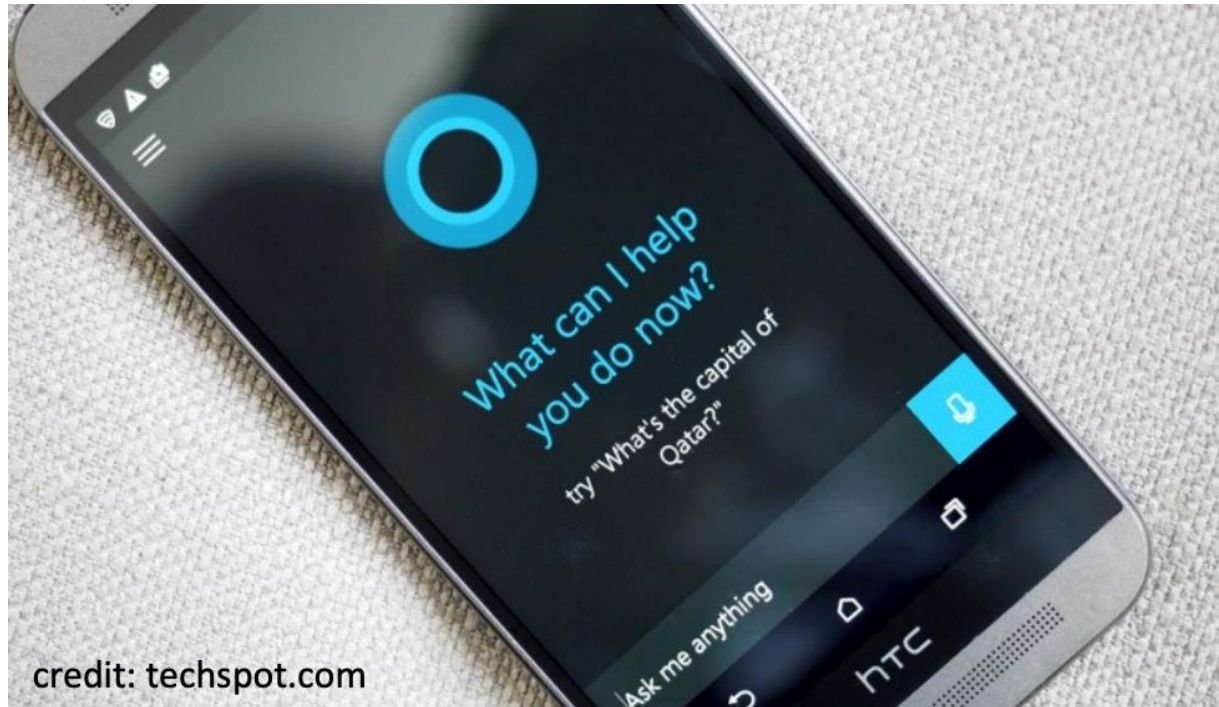
What is the weather in
Melbourne this Weekend?



Add *make booking reservation* to
my todo list

Digital personal assistant

More on natural language instruction



- Semantic parsing – understand tasks
- Entity linking – “my wife” = “Kellie” in the phone book

- Unstructured text to database entries

New York Times Co. named Russell T. Lewis, 45, president and general manager of its flagship New York Times newspaper, responsible for all business-side activities. He was executive vice president and deputy general manager. He succeeds Lance R. Primis, who in September was named president and chief operating officer of the parent.

Person	Company	Post	State
Russell T. Lewis	New York Times newspaper	president and general manager	start
Russell T. Lewis	New York Times newspaper	executive vice president	end
Lance R. Primis	New York Times Co.	president and CEO	start

Yoav Artzi: Natural language processing

Module I: Foundations of NLP

- **Week 1:** Information Retrieval Part 1
- **Week 2:** Information Retrieval Part 2
- **Week 3:** Text processing

Module II: Language Modeling & Representations

- **Week 4:** N-gram Language Models
- **Week 5:** Vector Embeddings and Sequence Labeling

Module III: Advanced NLP

- **Week 6:** Neural Networks for NLP (NNs, RNNs, and Neural LMs)
- **Week 7:** Transformers and Pretrained LMs
- **Week 8:** Large Language Models (LLMs)

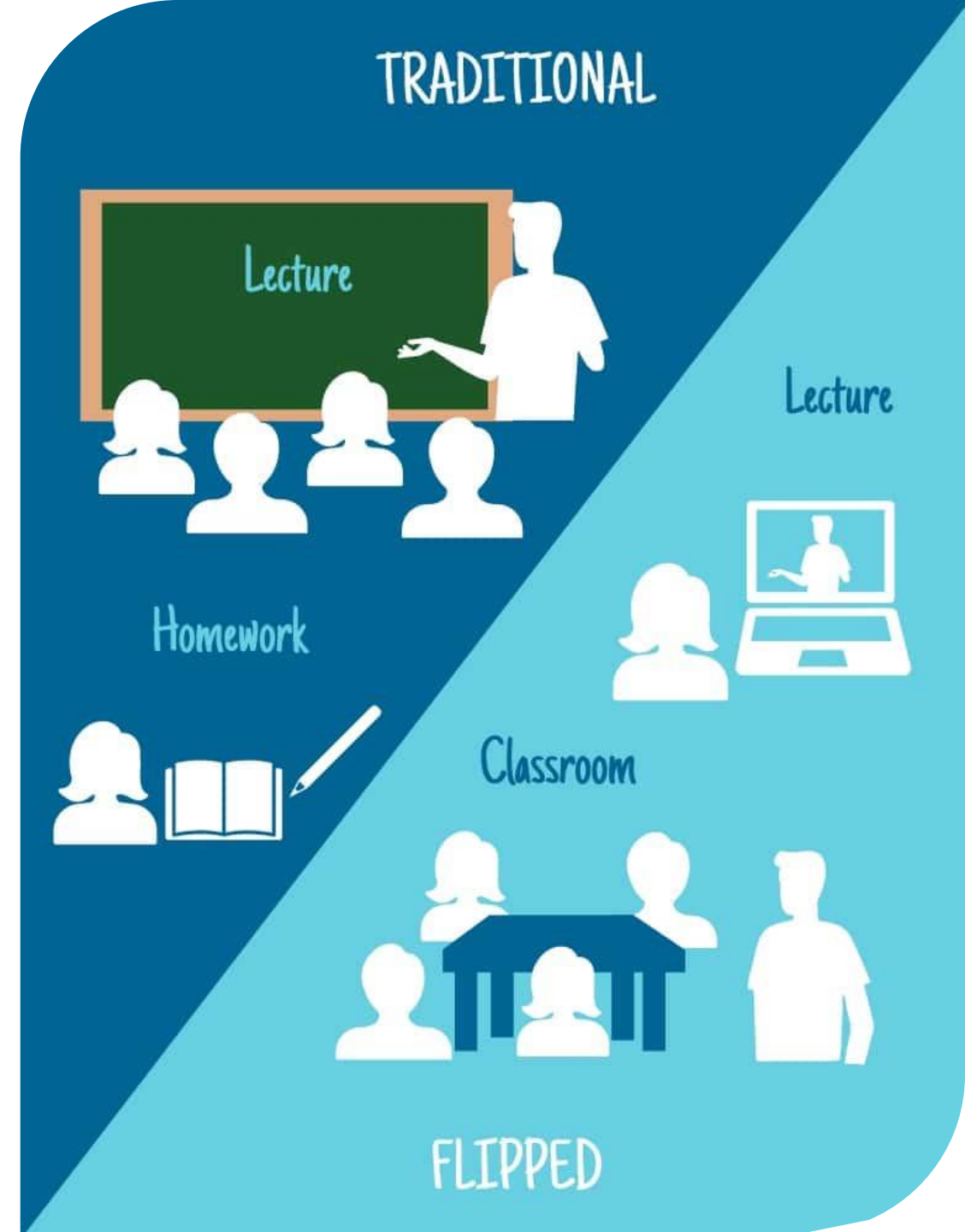
Module IV: Applications

- **Week 9:** Speech Processing & ASR
- **Week 10:** Dialogue Systems & Conversational AI
- **Week 11:** Wrap-up

Teaching: Engaging Learning Experience

What to Expect:

- **Active Learning:** Get ready for an interactive and hands-on learning experience.
- **Flipped Classroom Model:** We're flipping the traditional classroom! Pre-class materials will be provided for self-study before our sessions.
- **Your Role:** Come prepared to discuss, question, and apply concepts during class time.
- **Collaborative Environment:** Engage with your peers, share ideas, and actively participate.



- OnTrack learning portfolio 100% <https://ontrack.deakin.edu.au/>
 - An introduction video is available on DeakinCloud
- You set your target grade (P/C/D/HD)
 - Several tasks (tailored to your target grade) need to be completed regularly
 - Tasks have different due dates
- Weekly submissions are to get feedback
 - **Maximum two feedbacks. Please review carefully your assignment before submitting it!**
- Portfolio submission at the end of the trimester is **very important!**
 - **Assessments will be published gradually during the trimester!**
- **Please do not put assessments on public repositories (GitHub)!**



- **SIT330**

- Session 1:
 - Time: Monday 14:00-16:50
 - Location: Burwood B4.22
- Session 2:
 - Time: Friday 16:00-18:50
 - Location: Burwood B4.22
- Session 3:
 - Time: Thursday 18:00-19:50
 - Location: Online on MS Team



- **SIT770**

- Session 1:
 - Time: Friday 11:00-13:50
 - Location: Burwood T1.13
- Session 2:
 - Time: Tuesday 16:00-17:50
 - Location: Online on MS Teams
- Session 3:
 - Time: Tuesday 12:00-14:50
 - Location: Waurnd Pond IB4.202
- Session 4:
 - Time: Thursday 13:00-15:50
 - Location: Waurnd Pond IB4.202



Do not create a calendar meeting!
Weekly material will be gradually released!

- Please ask questions on the discussion forum to share with others and avoid redundant inquiries
- **Please use email only if necessary**
 - You must start your email subject with “[SIT330] or [SIT770]”
 - I will normally answer your email within 48 hrs (working days)
 - If you don’t receive a reply within 48 hrs, this often means that your question had been answered somewhere else before (e.g., in the class, forum, etc.)
 - If in doubt, check with me again during allocated contact time

This course requires you to
write substantial code in
Python to achieve a **C, D, or
HD grade!**



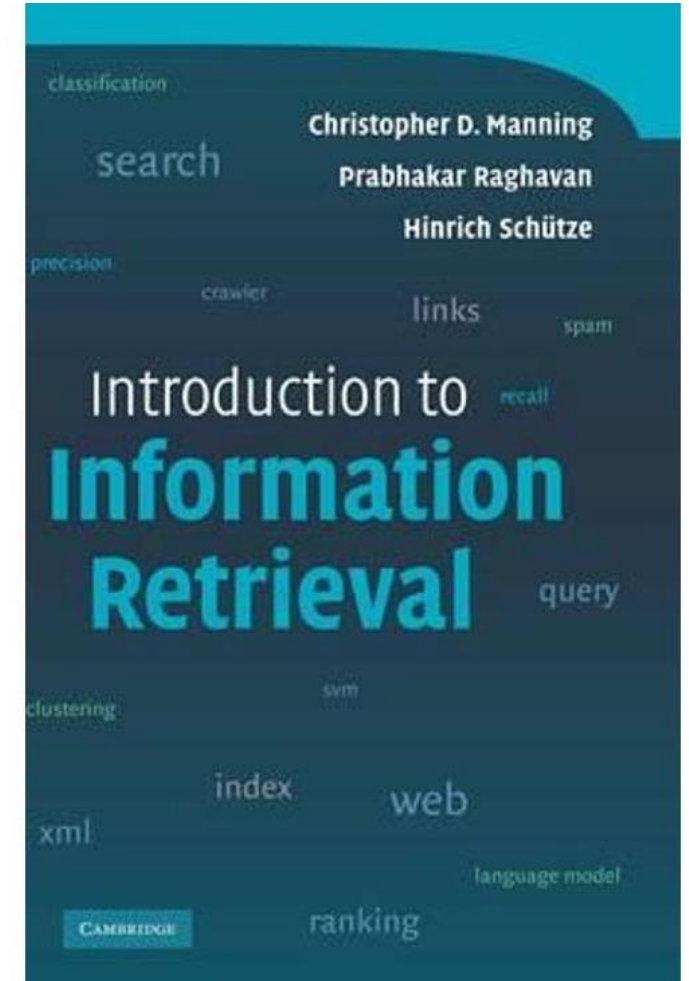
The unit does not teach
Python!

- Graduate Learning Outcomes (GLOs)
- Unit Learning Outcomes (ULOs)
- Teaching team contact details
- Learning activities
 - Class and workshop
- Reference resources
- Trimester plan
- Key dates

- **Introduction to Information Retrieval**
 - *Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze*
 - Draft chapters, April 1, 2009



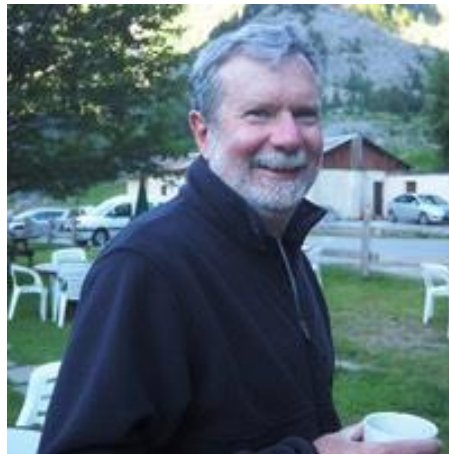
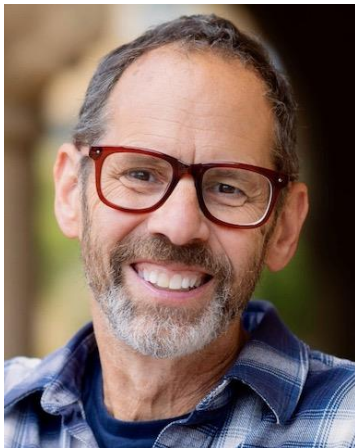
- <https://nlp.stanford.edu/IR-book/pdf/irbookonlinereading.pdf>



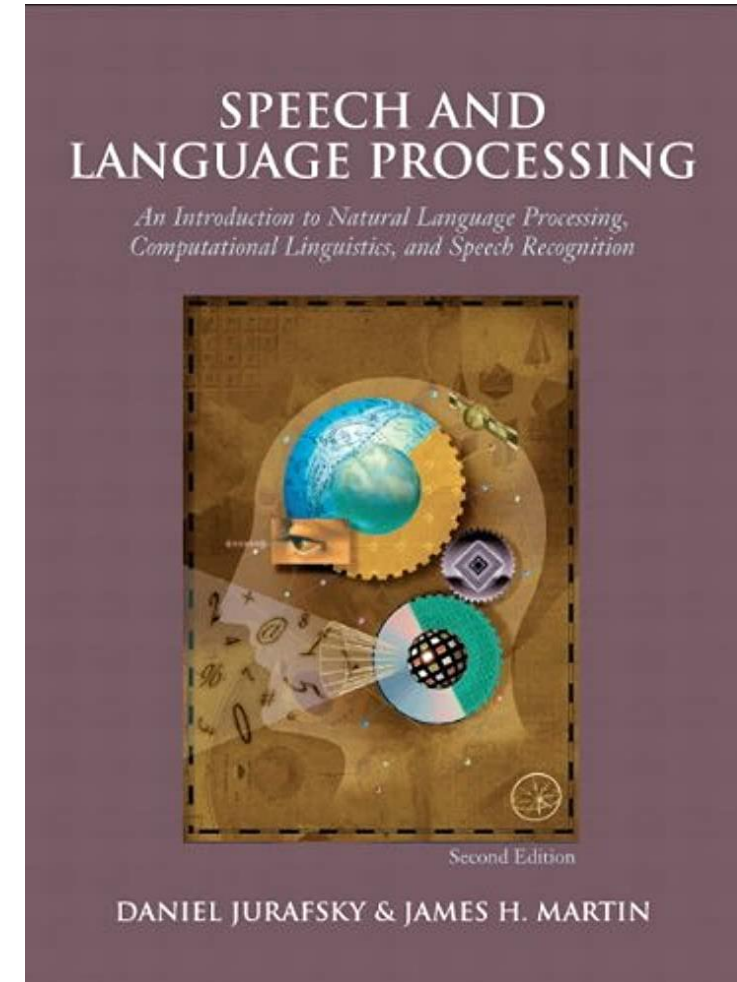
- **Speech and Language Processing (3rd ed. draft)**

- *Dan Jurafsky and James H. Martin*

- Draft chapters in progress, January 7, 2023



- <https://web.stanford.edu/~jurafsky/slp3/>



- SIT744- Deep Learning
 - Explain deep learning and its role in data science and AI

